INCH-POUND

MIL-C-50652/8B (AR)
12 December 1990

SUPERSEDING
MIL-C-50652/8A
23 April 1990

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548148
(CONTROL, SEARCH/TRACK, AND GAIN)

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for one type of circuit card assembly, hereafter referred to as assembly. The requirements of this specification, together with the general requirements of MIL-C-50652, comprise the complete requirements for the search/track, and gain control assembly used in Range Computer, CP-888/VPS-2, part of Radar Set AN/VPS-2 (see 6.1 of MIL-C-50652).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-50652 - Circuit Card Assemblies, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1285

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548148 - Search/Track and Gain Control
10548168 - Circuit, Integrated Single 8-Input NAND Gate
10548169 - Circuit, Integrated Quad Two Input NAND Gate
10548173 - Circuit, Integrated and Input J-K Flip Flop

PACKAGING DATA SHEET

PS10548148 - Packaging of Search/Track and Gain Control

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548148, all associated drawings, and with all requirements specified in applicable specifications and standards. The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in paragraph 3.3 MIL-C-50652.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the assembly.

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3.5 Performance. For the requirements of 3.5.3 through 3.5.4 logic input and output levels shall be as follows:

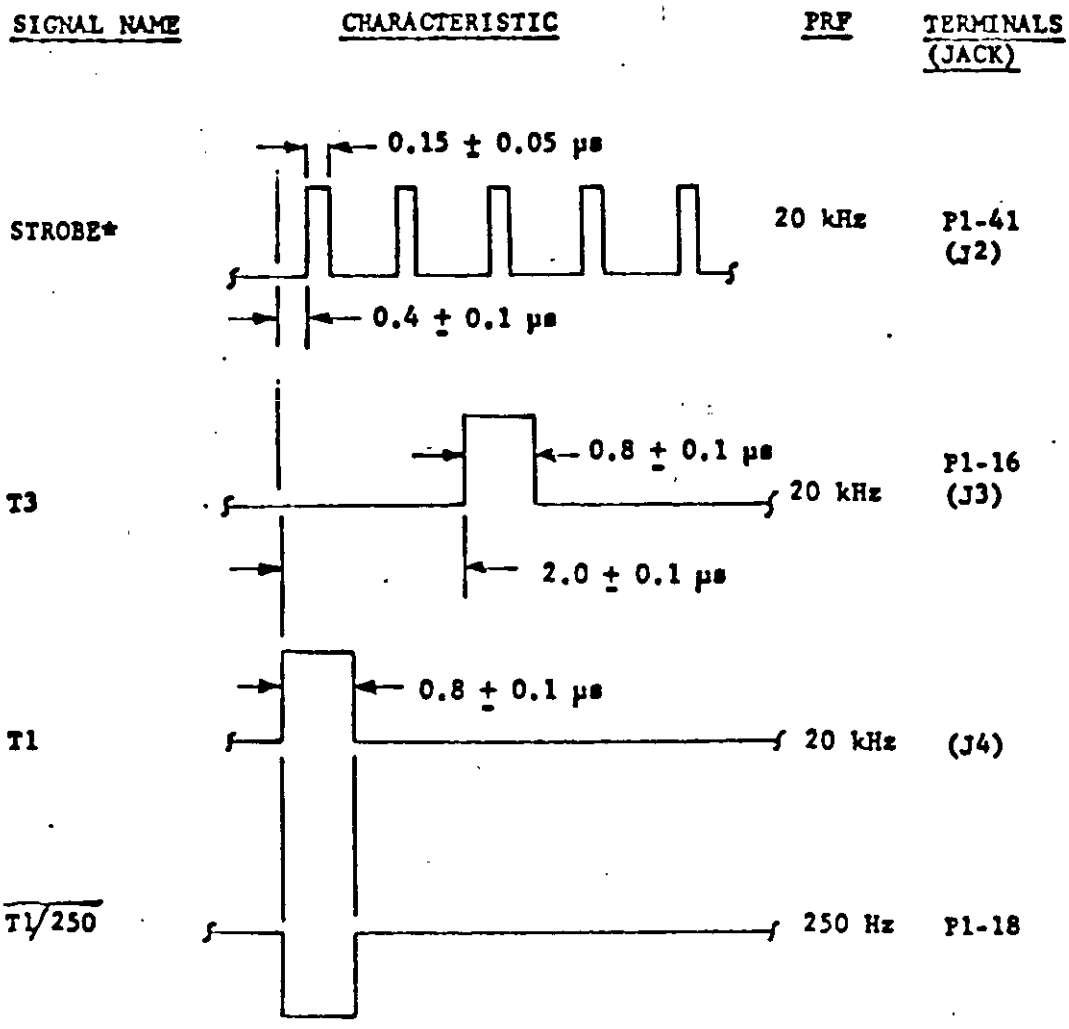
Logic 0: 0.0 ±0.5 volt direct current (Vdc) level
Logic 1: +3.5 ±1.0 Vdc level

3.5.1 Input power. The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ±0.3 Vdc, at 150 milliamperes (mA), regulated, with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signals. Input signals to the search/track and gain control assembly shall be STROBE, T3 and T1/250 as shown in Figure 1. A positive pulse shall be a change in direct current (dc) level from 0.0 ±0.5 V to +3.5 ±1.0 V and return to 0 V. A negative pulse shall be a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5V.

3.5.3 "TRACK" signal. With the logic level conditions, for P1-14, P1-20, P1-1, and P1-44, sequentially established as specified in Table I, steps 1 through 11, the track signal at P1-47 shall have the logic levels as specified in Table I. When the logic levels for P1-5 and P1-7 are simultaneously changed, the track output level shall change with the delays as specified in Table I.

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* STROBE signal is a train of 15 pulses at a 20 kHz PRF.

FIGURE 1. Input signal characteristics.

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TABLE I. Input-output signals

STEP	P1-14 (S1)	P1-20 (S2)	P1-1 (S3)	P1-44 (S5)	P1-7	P1-5 (S4)	P1-47 (TP3)	OUTPUT SIGNAL DELAY
	LOGIC LEVEL INPUT CONDITIONS							
1	0	1	1	1	0	0	0	
2					0	1	1	None
3					1	0	0	0.80 to 1.20 s
4	1	1	1	1	0	1	1	-
5					1	0	0	2.75 to 3.25 s
6	1	0	1	1	0	1	1	-
7					1	0	0	None
8	1	0	0	1	1	0	1	None
9	1	0	1	1	1	0	0	None
10	1	0	1	1	0	1	1	None
11	1	0	0	1	0	1	1	None
12	1	1	1	0	1	0	Fig.2	

3.5.3.1 TRACK signal test. With the input logic levels as specified in Table I, step 12, the track output signal shall be as specified in Figure 2.

3.5.4 Gain signals. For 3.5.4.1 through 3.5.4.3 the input logic levels for P1-20, P1-14, P1-1, and P1-44 shall be as specified for Table II, step 1.

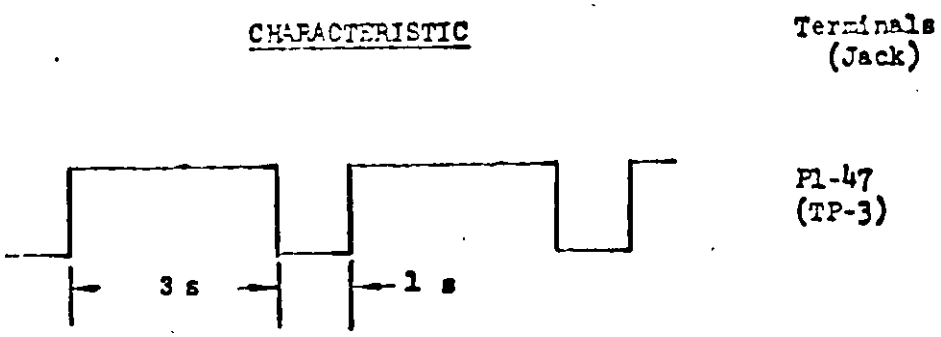


FIGURE 2. Output signal characteristic.

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TABLE II. Input signals

INPUT LOGIC LEVELS		
STEP	P1-7	P1-5 (S4)
1	1	0
2	0	1

3.5.4.1 G1 signal. The G1 output signal at P1-11 shall change from logic 1 to logic 0 between 0.125 and 0.375 s after the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, step 1 to step 2. The G1 signal shall change from logic 0 to logic 1 between 0.80 and 1.20 s when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, step 2 to step 1.

3.5.4.2 G2 signal. The G2 output signal at P1-15 shall be a single positive pulse between 0.125 and 0.375 s wide which occurs when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, step 1 to step 2.

3.5.4.3 G3 signal. The G3 output signal at P1-13 shall change from logic 0 to logic 1 between 0.25 and 0.75 s after the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, step 1 to step 2. The G3 signal shall change from logic 1 to logic 0 immediately when the input logic levels for P1-7 and P1-5 are simultaneously changed from the conditions of Table II, step 2 to step 1.

3.6 First article. As specified in paragraph 3.6 of MIL-C-50652.

3.7 Soldering. As specified in paragraph 3.7 of MIL-C-50652.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. As specified in paragraph 4.1 of MIL-C-50652.

4.2 Classification of inspections. As specified in paragraph of 4.2 MIL-C-50652.

4.3 First article inspection. As specified in paragraph 4.3 of MIL-C-50652.

4.4 Quality conformance inspection. As specified in paragraph 4.4 of MIL-C-50652.

4.5. Special Sampling. As specified in MIL-C-50652.

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4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. As specified in paragraph 4.6.1 of MIL-C-50652.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. As specified in paragraph 4.6.2.1 of MIL-C-50652.

4.6.2.2 Contractor design. As specified in paragraph 4.6.2.2 of MIL-C-50652.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE III. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated; ac ripple less than 50 mV peak-to-peak; capacity 250 mA.
2. Pulse generator	Train of 15 pulses at 20 KHz externally synchronized. Amplitude 3.5 ±1.0 V peak-to-peak into 50-ohm load.
3. Pulse generator	Two separate output pulses, 20 KHz PRF, controllable in separation, amplitude 3.5 ±1.0 V peak-to-peak into 50-ohm load.
4. Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less, sweep period 50 ns per cm or greater, input resistance greater than 8 megohms, input capacitance less than 25 PF.
5. Test fixture	As specified in Figure 3. See Table IV for test fixture parts list.

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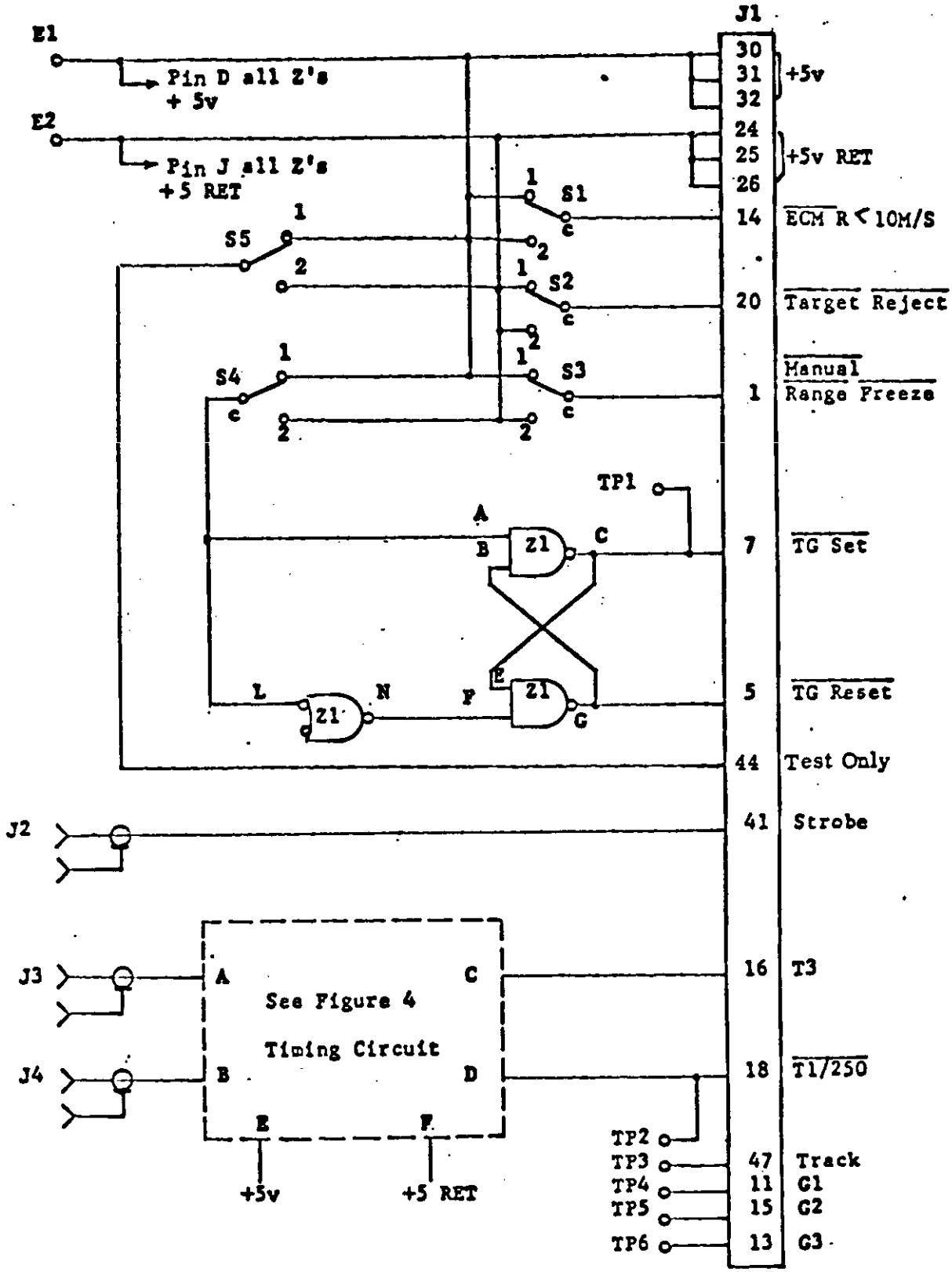


FIGURE 3. Test fixture setup.

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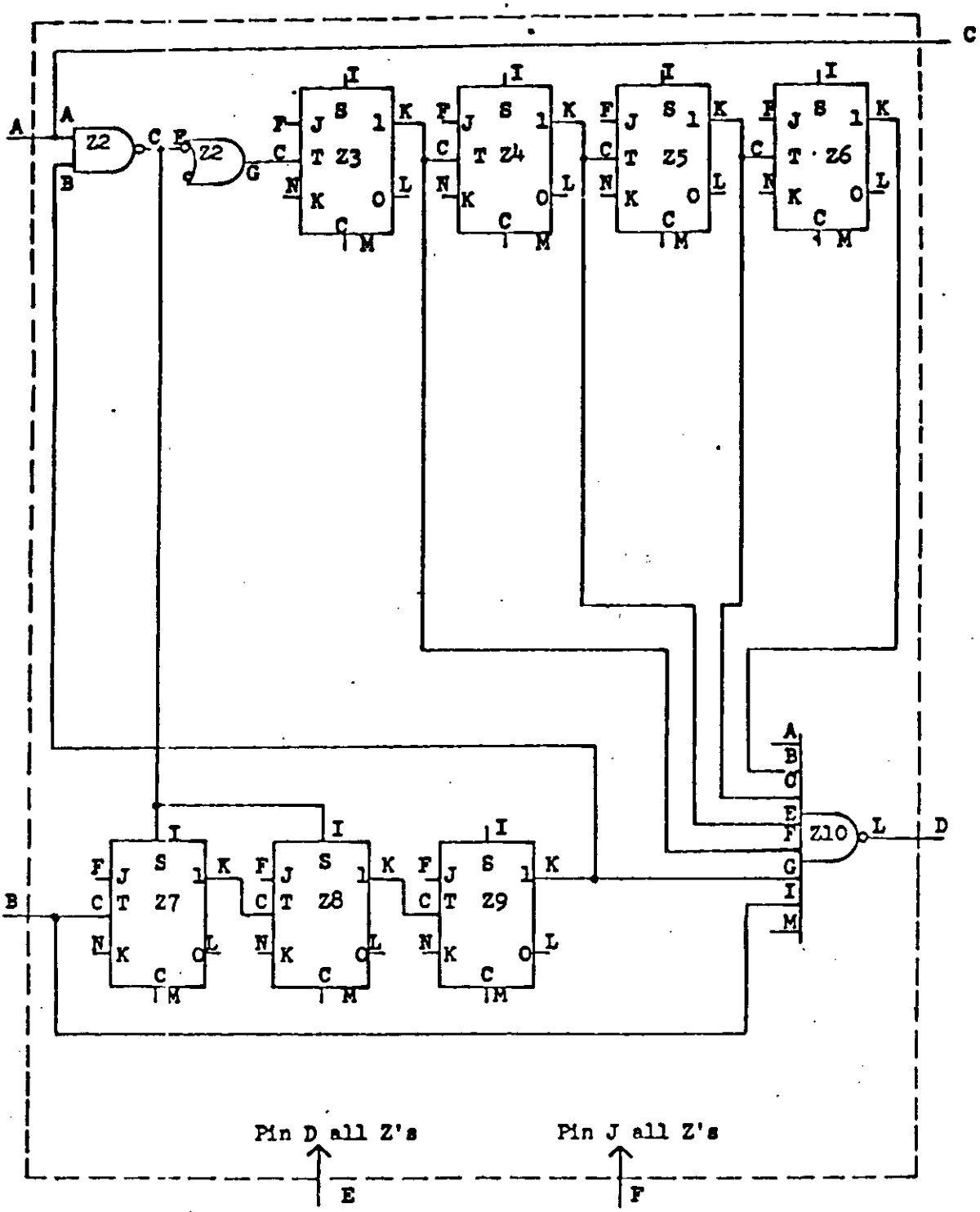


FIGURE 4. Test fixture timing circuit.

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TABLE IV. Test fixture parts list

DESIGNATION	
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector, any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
J4	Same as J2.
S1 thru S5	Switch SPDT.
TP1 thru TP6	Tip jack, H. H. Smith 1501 or equivalent.
Z1 and Z2	Integrated Circuit, 10548169 (RX3333, Raytheon) or equivalent.
Z3 thru Z9	Integrated Circuit, 10548173 (RX3371, Raytheon) or equivalent.
Z10	Integrated Circuit, 10548168 (RX3369, Raytheon) or eqivalent.

4.7 Methods of inspection.

4.7.1 Environmental. As specified in paragraph 4.7.1 of MIL-C-50652.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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4.7.3 Performance. Set up equipment as shown in Figure 5. Use oscilloscope (Table III, item 4), for observation of input and output signals specified in 4.7.3.2 through 4.7.3.4. For observation of delay times use oscilloscope (Table III, item 4) in single sweep mode of operation using TP1 trigger for synchronization. It will be necessary to reset the oscilloscope after each delay test step is initiated. Switches S1 through S3 and S5, on the test fixture (Table III, item 5) are used to supply the logic level input signals specified in Table I. These switches supply a logic 1 in position 1 and a logic 0 in position 2. Switch S5 is used to supply the logic levels for Tables I and II. This switch supplies the logic levels of Table II, step 1, when in position 2 and the levels of step 2 when in position 1.

4.7.3.1 Input power. Using power supply (Table III, item 1) apply input power as specified in 3.5.1.

4.7.3.2 Input signals. Use pulse generators (Table III, items 2 and item 3) to supply the STROBE T3, and T1 signals as shown in Figure 1. Connect these signals to J2, J3 and J4 respectively. At TP2, check the T1/250 input signal for the requirements of Figure 1.

4.7.3.3 TRACK signal. Using switches S1 through S5 to establish the logic levels as specified in 3.5.3, observe, at TP3, the track output signal as specified in 3.5.3.

4.7.3.3.1 TRACK signal test. Using switches S1 through S5 to establish the logic levels of Table I, step 12, observe, at TP3, the track signal shown in Figure 2.

4.7.3.4 Gain signals. Using switches S1 through S3 and S5 establish the logic level conditions of Table 1, step 1.

4.7.3.4.1 G1 signal. While using switch S4 to establish the logic level conditions of Table II, observe at TP4, the G1 signal as specified in 3.5.4.1.

4.7.3.4.2 G2 signal. While using switch S4 to establish the logic level conditions of Table II, observe, at TP5, the G2 signal as specified in 3.5.4.2.

4.7.3.4.3 G3 signal. While using switch S4 to establish the logic level conditions of Table II, observe at TP6, the G3 signal as specified in 3.5.4.3.

4.8 Soldering. Test as specified in paragraph 4.8 of MIL-C-50652.

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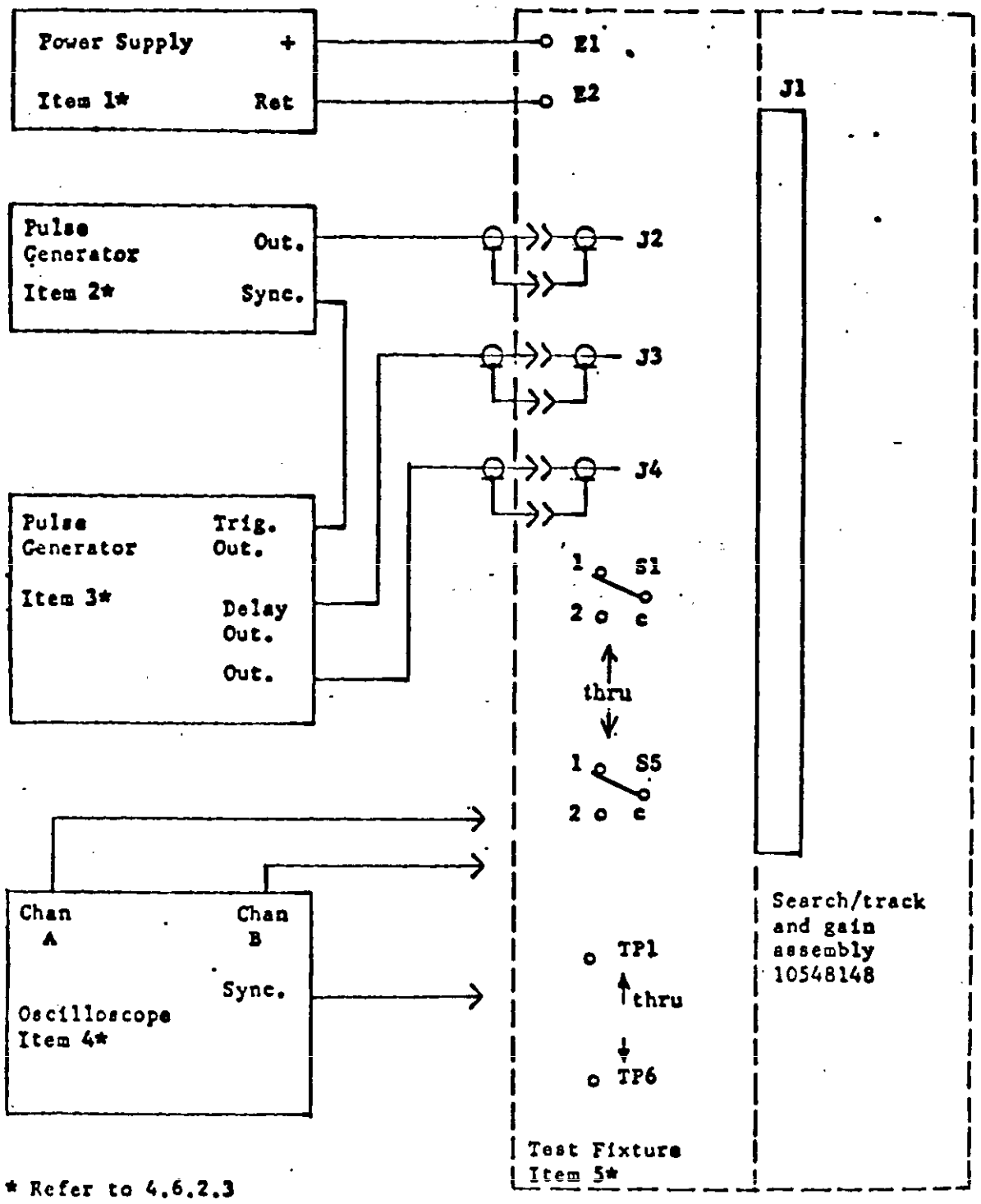


FIGURE 5. Search/track and gain control test setup.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS10548148. The level of protection whall be as specified in the procurement document.

6. NCTES. As specified in section 6 (Notes) of MIL-C-50652.

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